

Physics-Informed Bayesian Wildfire Exposure Ranking for Transmission-Line Segments Using Public California Data

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Abstract

Extreme wildfire decision support for linear infrastructure often requires ranking assets before complete information about future fire weather, fuel conditions, and fire spread is available. This paper develops a reproducible public-data pipeline for ranking California transmission-line segments by external wildfire exposure. Transmission lines are treated as receptor assets, not ignition sources. The main validation label excludes CAL FIRE electrical power ignitions to avoid conflating externally caused wildfire exposure with line-caused fire occurrence. Using CEC transmission lines, CAL FIRE perimeters, gridMET fire-weather summaries, LANDFIRE fuel structure, and USGS terrain data, we construct a 2017–2023 segment-year panel for the Northern Sierra and Southern Cascades. A physics-informed Bayesian logistic model is compared with a calibrated deterministic score and machine-learning baselines under a 2022–2023 temporal holdout. Bayesian posterior ranking improves rare-event decision concentration relative to the deterministic score and yields decision-facing quantities including posterior exceedance probability and posterior top-decile rank probability. Stan MCMC pilot results agree closely with the Laplace posterior prototype, supporting the use of posterior decision metrics for extreme-disaster triage.

1 Introduction

Wildfire risk decisions for transmission corridors are made under extreme uncertainty. Operational risk scores are often deterministic: they compress weather, fuel, terrain, and asset context into a single ordered list. Such scores can be transparent, but they do not directly represent uncertainty in the exceedance or ranking decisions that matter for rare disasters.

This study asks whether fully public data can support a reproducible, physics-informed Bayesian exposure-ranking pipeline for externally exposed linear assets. The key distinction is that the study estimates external wildfire exposure of transmission-line segments, not powerline-caused ignition, equipment failure, or outage probability.

The contributions are:

1. a public-data segment-year exposure panel for California transmission line segments;
2. cause-screened external wildfire exposure labels that exclude electrical-power fire causes from the main validation target;
3. a physics-informed Bayesian ranking model with posterior exceedance and posterior top-decile rank probabilities;
4. temporal-holdout validation against deterministic and machine-learning baselines; and

5. a Stan MCMC implementation path that reproduces the Laplace posterior ranking prototype.

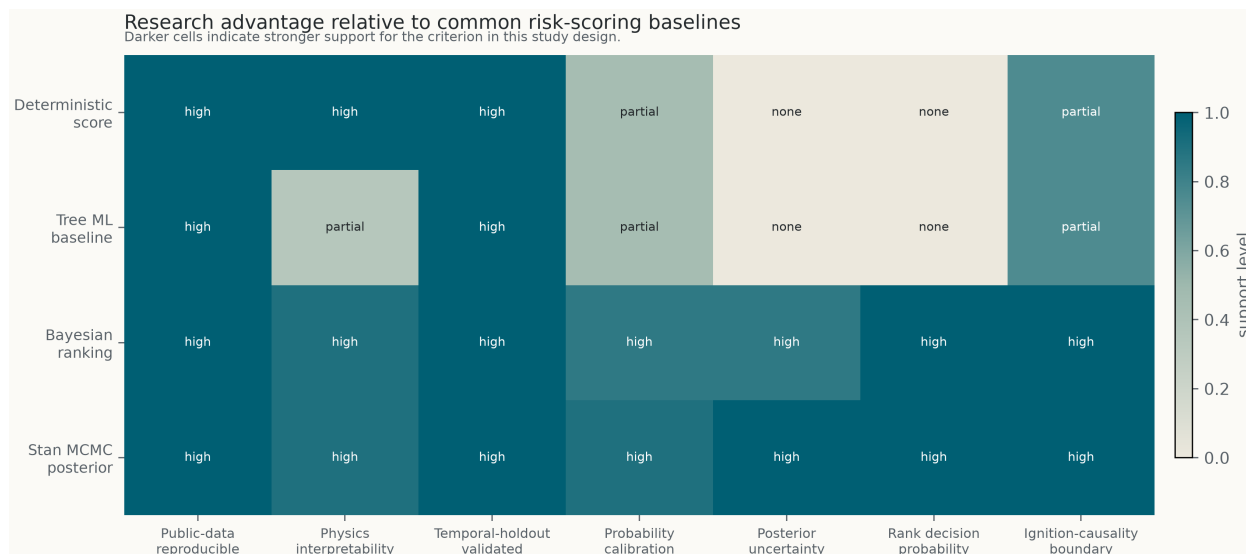


Figure 1: Research advantage relative to deterministic risk scores and flexible machine-learning baselines. The contribution is strongest where public reproducibility, physical interpretability, temporal validation, posterior uncertainty, and rank-decision probabilities are needed together.

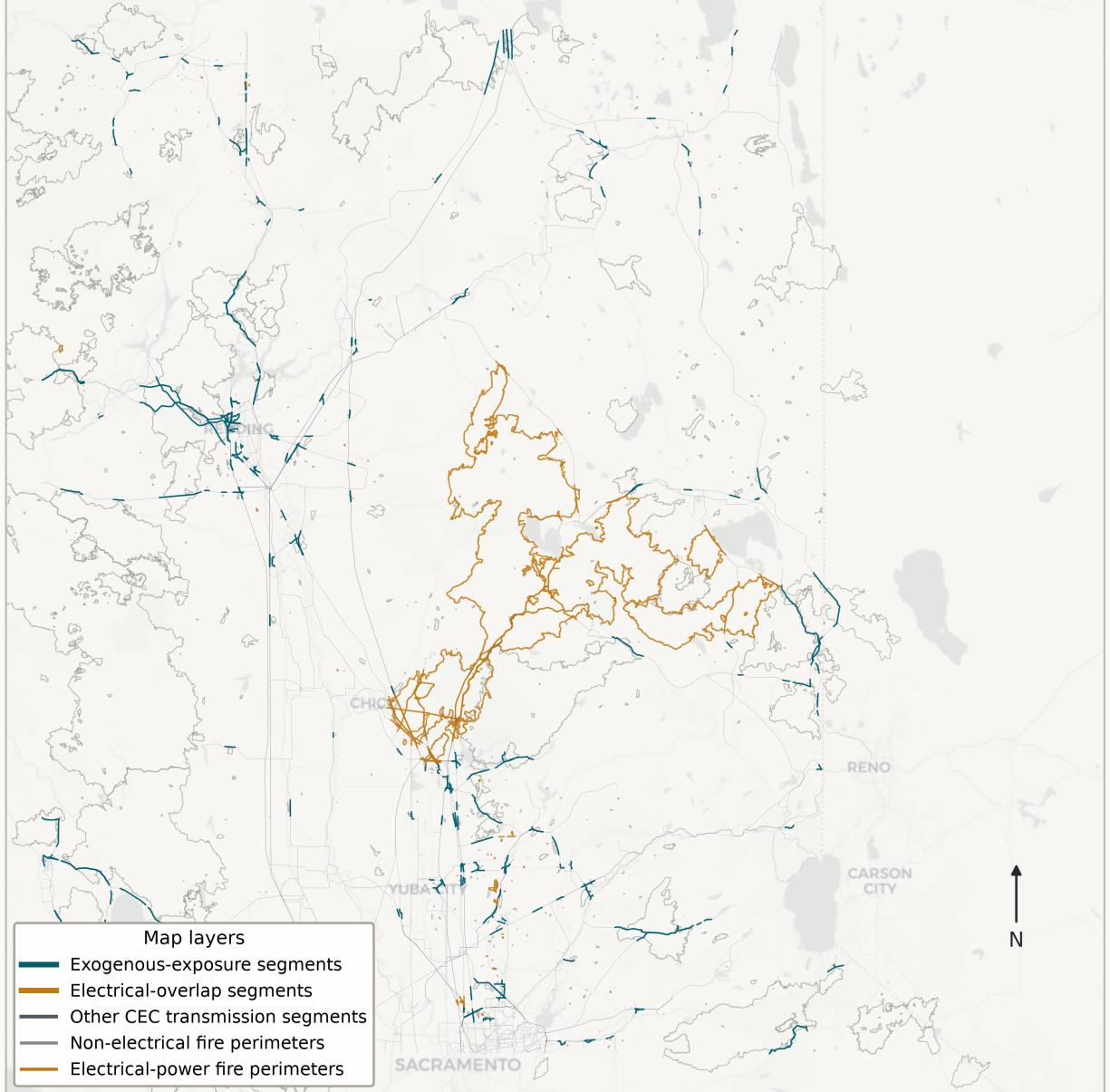
2 Study Area and Data

2.1 Study Area

The case study focuses on the Northern Sierra and Southern Cascades in California. The decision units are approximately 1 km transmission-line segments. The analysis panel spans 2017–2023 and uses segment-year rows so that annual fire labels and annual fire-weather summaries are aligned.

Transmission-line wildfire exposure labels in Northern California

1 km CEC line segments, 1 km buffers, CAL FIRE perimeters 2017-2023



Sources: CEC transmission lines; CAL FIRE FRAP fire perimeters; CartoDB Positron basemap.

Figure 2: Study area, transmission-line segments, and annual wildfire exposure labels. Transmission lines are treated as receptor assets for external wildfire exposure.

2.2 Public Data Sources

The pipeline uses California Energy Commission transmission lines, CAL FIRE historical fire perimeters and cause fields, gridMET weather and fire-danger variables, USGS 3DEP elevation, and LANDFIRE fuel and canopy layers. The main processed table contains 11,255 line segments and 78,785 segment-year observations.

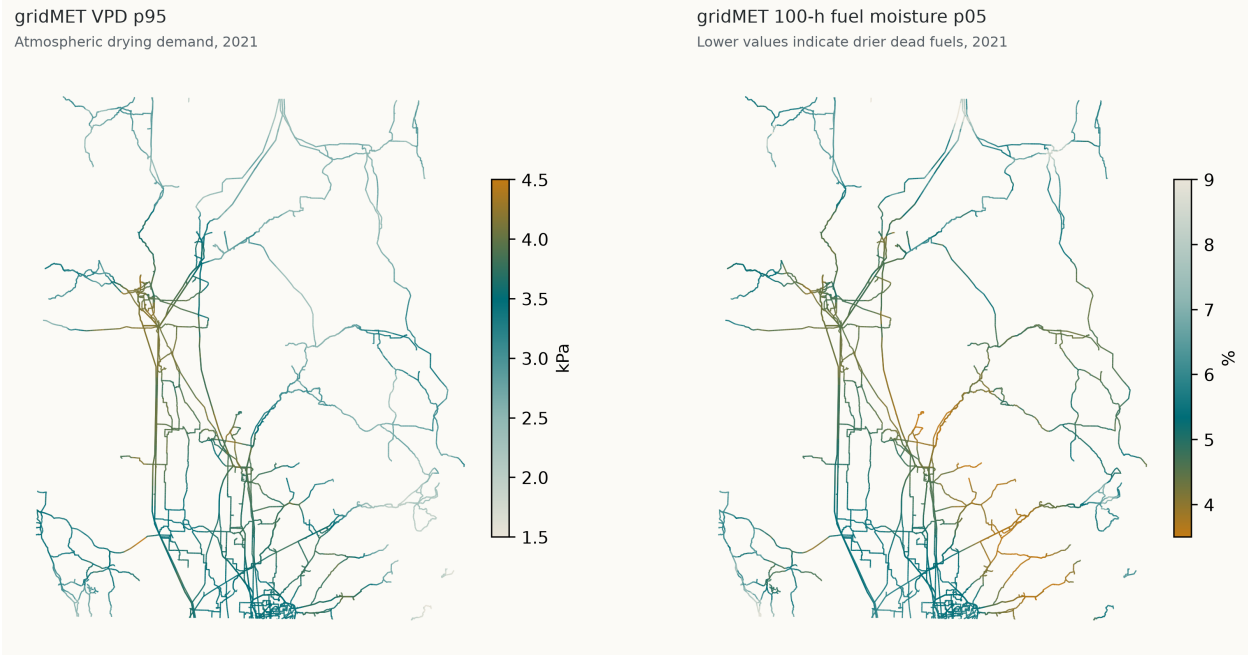


Figure 3: Public environmental covariates used to construct dynamic weather, terrain, and fuel-structure predictors. (a) Dynamic gridMET covariates summarized to transmission-line segments for atmospheric drying demand and dead-fuel moisture.

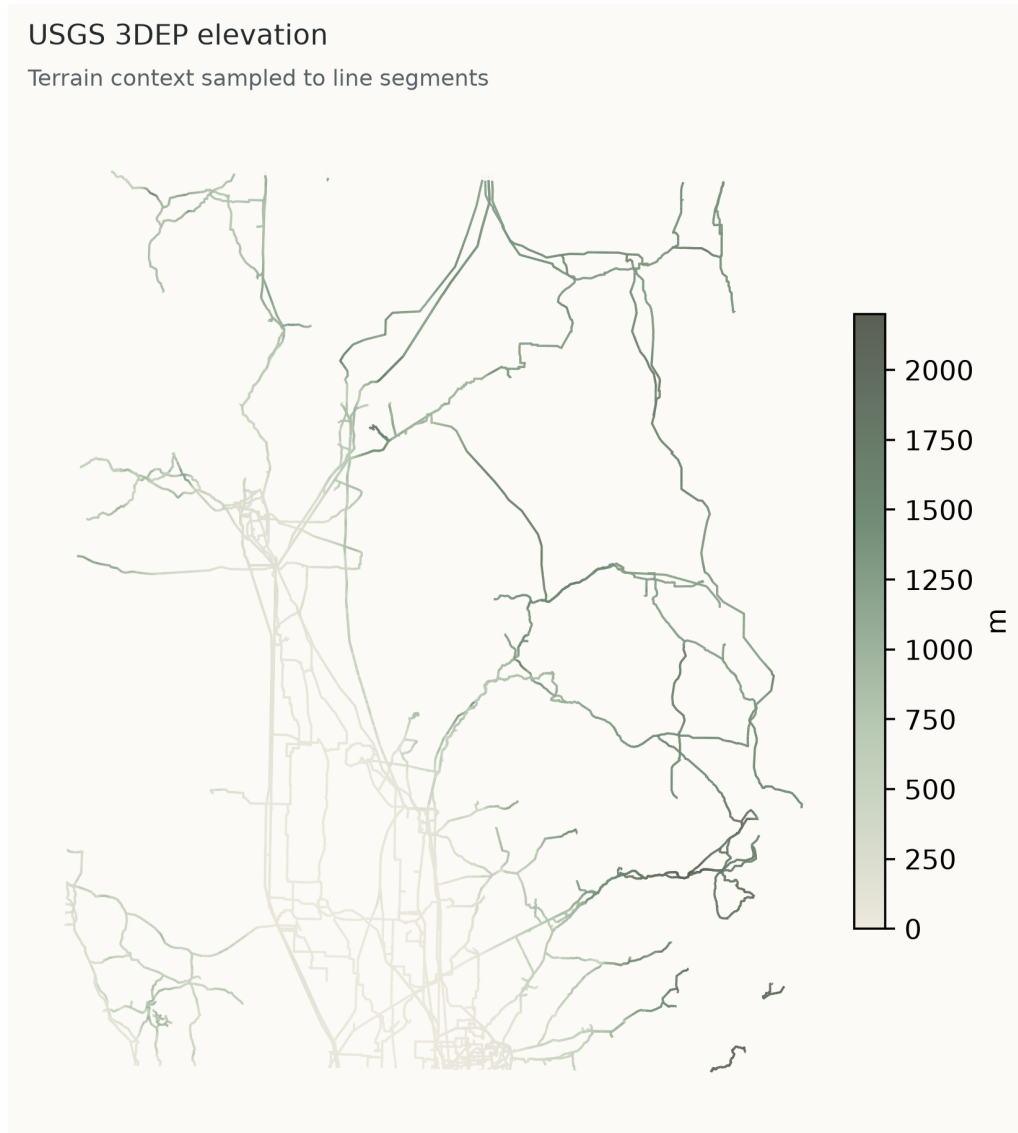


Figure 3: Public environmental covariates used to construct dynamic weather, terrain, and fuel-structure predictors. (b) USGS 3DEP elevation sampled to transmission-line segments as static terrain context.

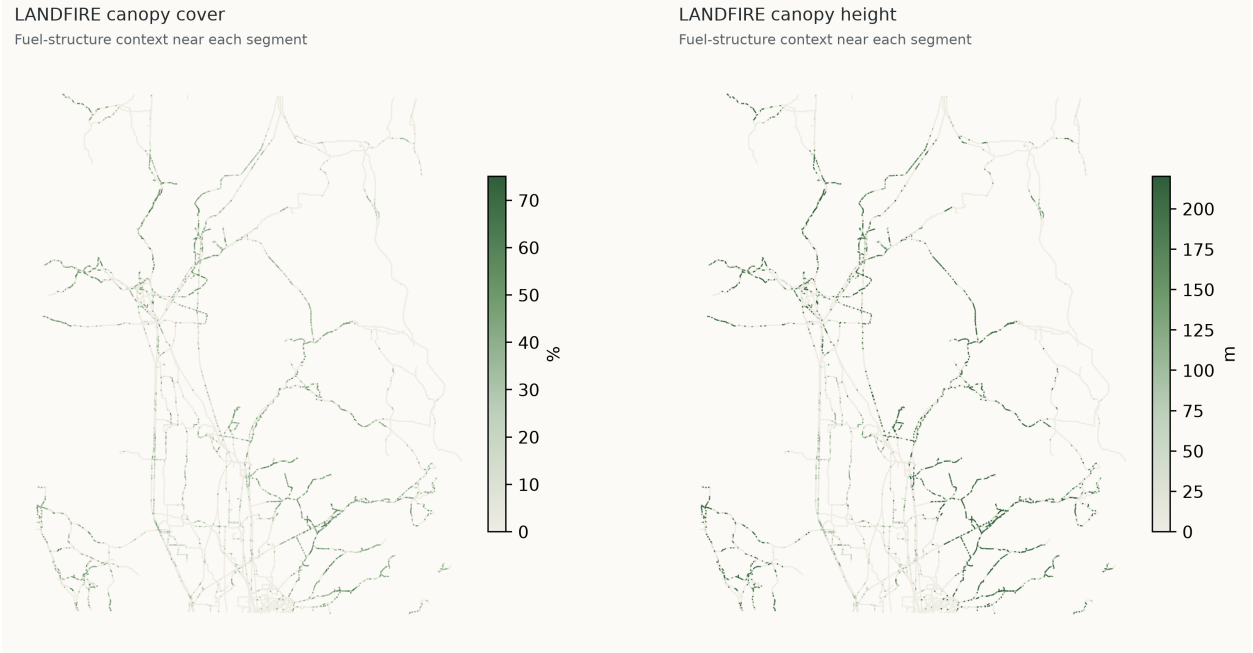


Figure 3: Public environmental covariates used to construct dynamic weather, terrain, and fuel-structure predictors. (c) LANDFIRE canopy cover and canopy height summarized to line segments as static fuel-structure context.

3 Label Construction

Let i index line segments and t index years. The main label is

$$y_{it}^{exo} = \mathbb{I}\{\text{buffer}_i \cap \text{fire perimeter}_t \neq \emptyset, \text{CAUSE} \neq 11\}, \quad (1)$$

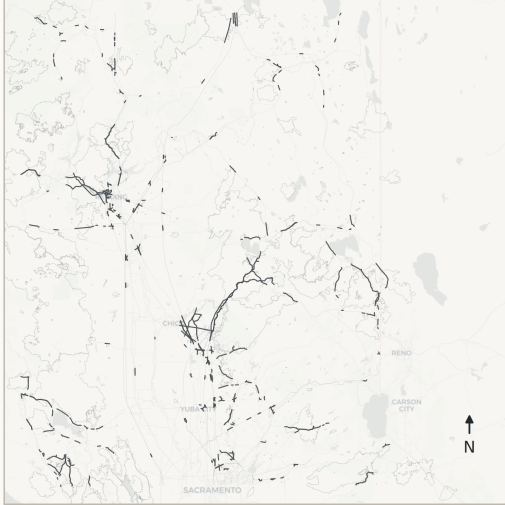
where CAL FIRE cause code 11 corresponds to electrical power. This exclusion is used to keep the validation target aligned with external wildfire exposure rather than line-caused ignition.

The processed labels include all-fire exposure, exogenous exposure, strict exposure, and electrical-power overlap. Across 2017–2023, the panel contains 3,187 all-fire exposure positives, 2,331 exogenous exposure positives, 2,075 strict exposure positives, and 885 electrical-power overlap positives.

Cause-screened segment-year exposure labels

Labels are parallel diagnostic definitions, not mutually exclusive classes.

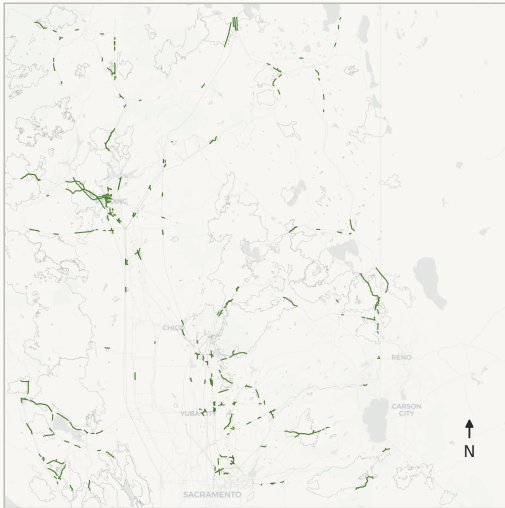
A. All fire exposure
3,187 positive segment-year rows



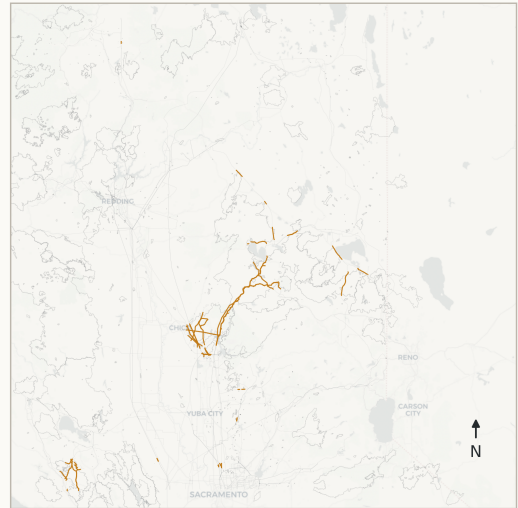
B. Exogenous exposure (CAUSE != 11)
2,331 positive segment-year rows



C. Strict exogenous (CAUSE not in {11, 2})
2,075 positive segment-year rows



D. Electrical-power overlap diagnostic
885 positive segment-year rows



— All fires — Exogenous — Strict exogenous — Electrical diagnostic

Figure 4: Cause-screened exposure labels. Electrical-power fire overlap is separated from the main exogenous exposure label to preserve the receptor asset interpretation.

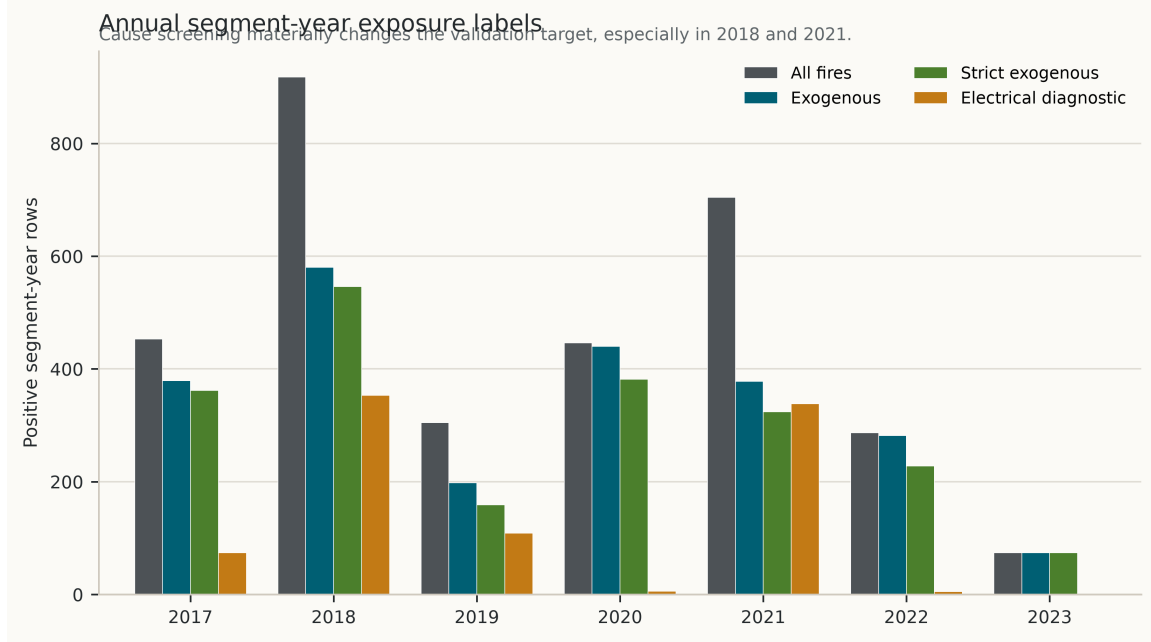


Figure 5: Annual segment-year label counts used for temporal validation.

4 Methods

4.1 Physics-Informed Feature Core

The selected feature core combines dynamic gridMET dryness and wind summaries, LANDFIRE fuel-structure variables, terrain, and engineered physics indices:

VPD, ERC, BI, wind speed, 100-hour and 1000-hour fuel moisture, minimum relative humidity, precipitation, elevation, canopy cover, canopy height, canopy base height, canopy bulk density, dryness index, fuel-structure index, and dryness-by-fuel interaction.

Bayesian exposure-ranking workflow and mathematical decision layer

The asset is a receptor interface; the posterior ranks external wildfire exposure, not line-caused ignition.

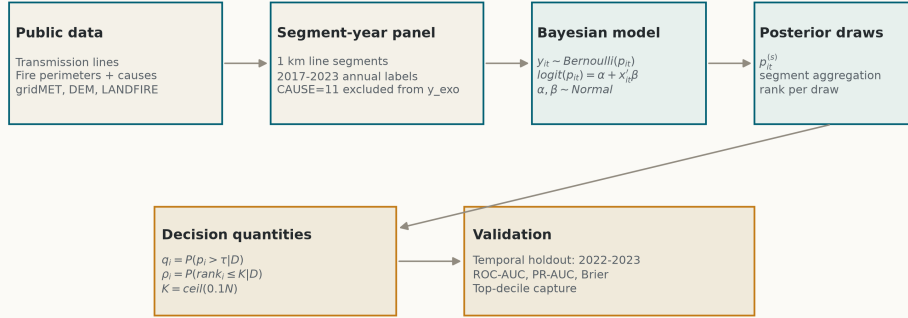


Figure 6: Bayesian exposure-ranking workflow and mathematical decision layer. Public geospatial data are converted to a segment-year panel, fitted with a Bayesian exposure model, and summarized as posterior exceedance and posterior rank probabilities.

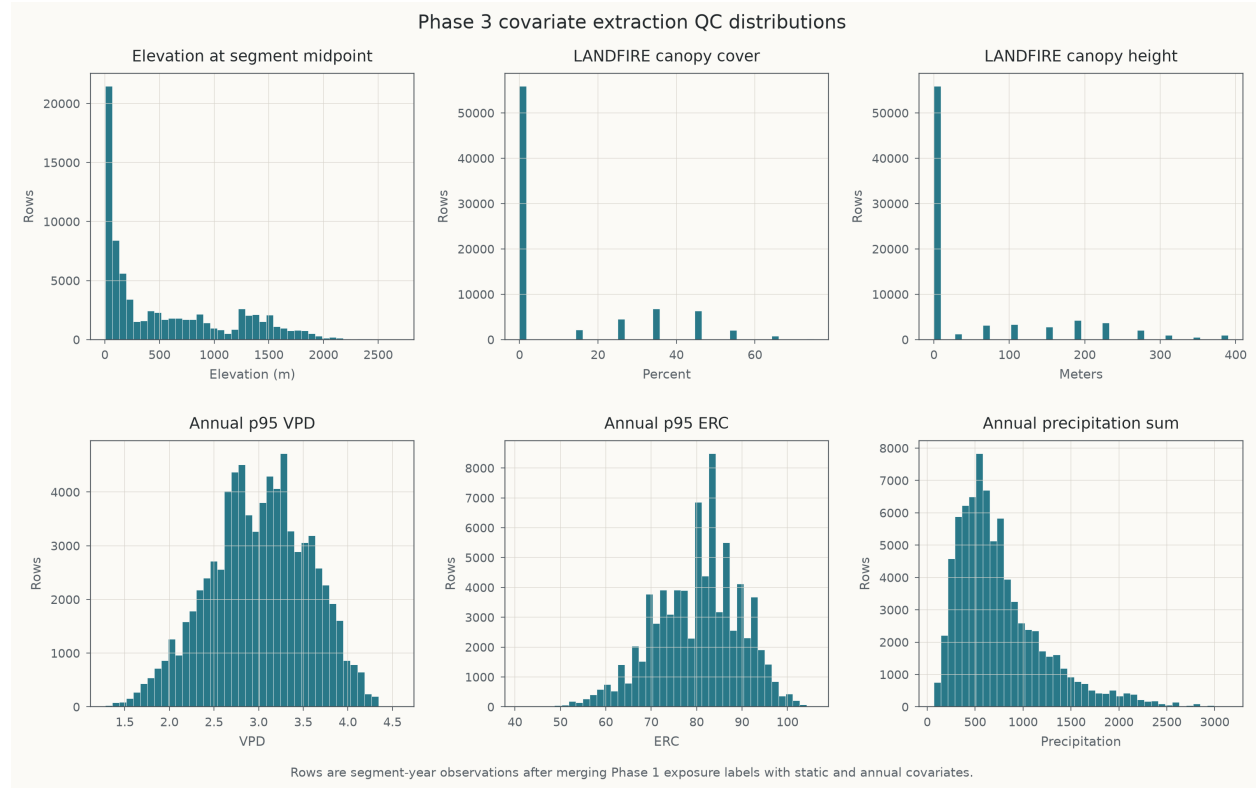


Figure 7: Quality-control distributions for model covariates.

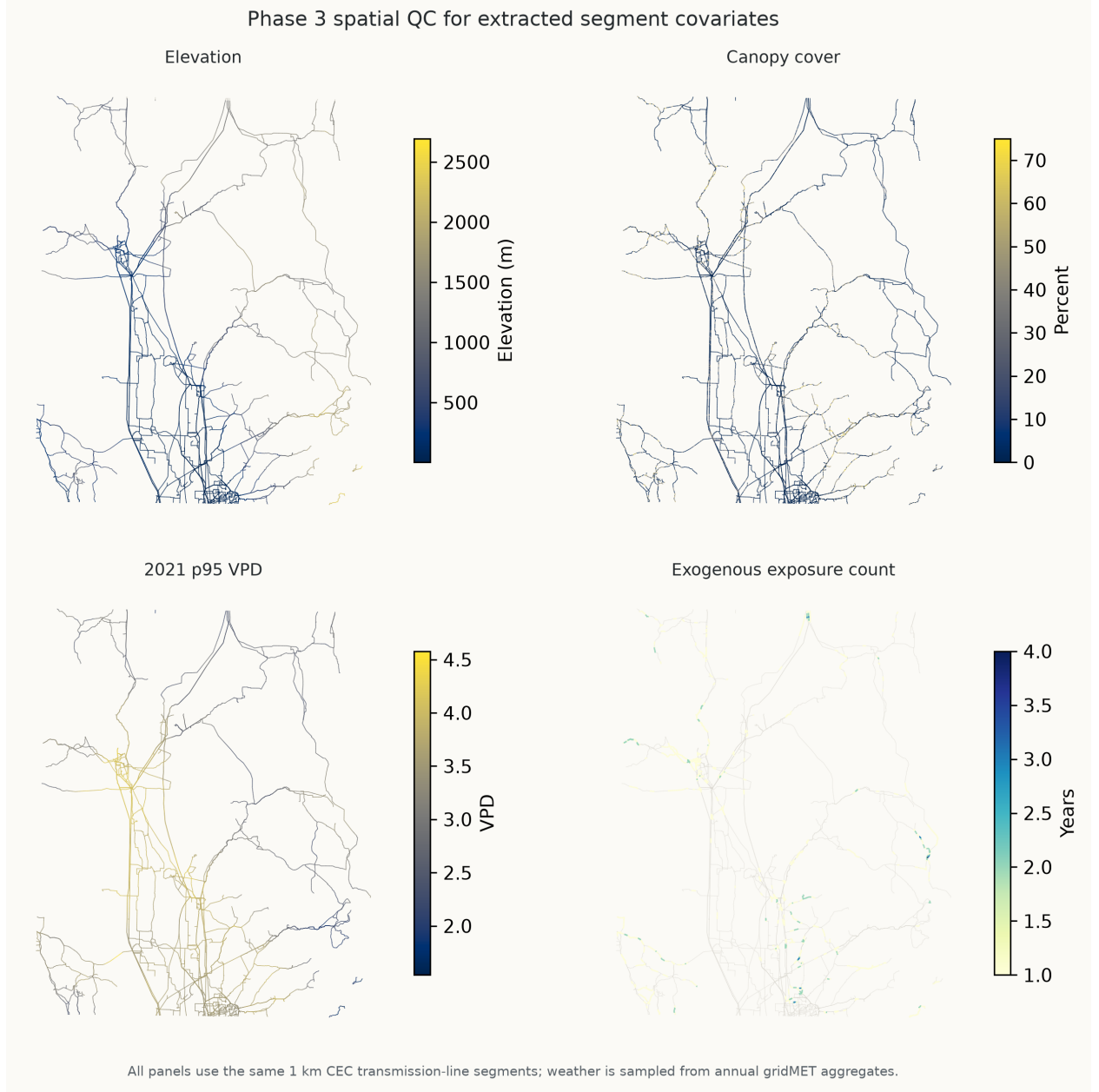


Figure 8: Spatial quality-control maps for static and dynamic covariates.

4.2 Deterministic Baseline

The deterministic comparator is a calibrated physics score based on dryness, fuel structure, and elevation. It is used as a transparent benchmark for ranking, not as the final decision model. Let $z_{k,it}$ denote standardized covariates estimated using the training period. The deterministic baseline can be written as

$$s_{it}^{det} = \text{clip}_{[0,1]} \left[a_1 z_{it}^{dry} + a_2 z_i^{fuel} + a_3 z_i^{elev} \right], \quad (2)$$

where the score is subsequently calibrated to a probability scale using a regularized logistic calibration. This baseline yields a single ranking but no posterior distribution over ranks.

4.3 Bayesian Exposure Model

The main Bayesian model is a weakly regularized logistic exposure model. For segment i and year t :

$$y_{it}^{exo} \sim \text{Bernoulli}(p_{it}), \quad (3)$$

$$\eta_{it} = \alpha + \mathbf{x}_{it}^\top \boldsymbol{\beta}, \quad (4)$$

$$p_{it} = \text{logit}^{-1}(\eta_{it}), \quad (5)$$

$$\alpha \sim \mathcal{N}(0, 5^2), \quad (6)$$

$$\beta_k \sim \mathcal{N}(0, 2^2). \quad (7)$$

The posterior is

$$p(\alpha, \boldsymbol{\beta} \mid D) \propto \prod_{(i,t) \in D} p_{it}^{y_{it}} (1 - p_{it})^{1-y_{it}} p(\alpha) \prod_k p(\beta_k). \quad (8)$$

The implementation is first fit using a Laplace posterior approximation around the posterior mode and then checked with a Stan MCMC pilot.

Posterior draws are indexed by $s = 1, \dots, S$:

$$(\alpha^{(s)}, \boldsymbol{\beta}^{(s)}) \sim p(\alpha, \boldsymbol{\beta} \mid D), \quad (9)$$

$$p_{it}^{(s)} = \text{logit}^{-1} \left(\alpha^{(s)} + \mathbf{x}_{it}^\top \boldsymbol{\beta}^{(s)} \right). \quad (10)$$

For segment-level decisions over a holdout period $T^\star$, posterior segment exposure is aggregated as

$$\bar{p}_i^{(s)} = \frac{1}{|T^\star|} \sum_{t \in T^\star} p_{it}^{(s)}. \quad (11)$$

The decision quantities are:

$$q_i = P(\bar{p}_i > \tau \mid D) \approx \frac{1}{S} \sum_{s=1}^S \mathbb{I}\{\bar{p}_i^{(s)} > \tau\}, \quad (12)$$

$$\rho_i = P(\text{rank}(\bar{p}_i) \leq K \mid D) \approx \frac{1}{S} \sum_{s=1}^S \mathbb{I}\{\text{rank}(\bar{p}_i^{(s)}) \leq K\}, \quad (13)$$

where τ is the training prevalence and K is the top-decile segment count.

4.4 Validation Design

Models are trained on 2017–2021 and evaluated on 2022–2023. The temporal holdout avoids leakage from repeated segment observations across years. Metrics include ROC-AUC, PR-AUC, Brier score, row-level top-decile capture, and segment-level top-decile capture.

5 Results

5.1 Deterministic Versus Bayesian Ranking

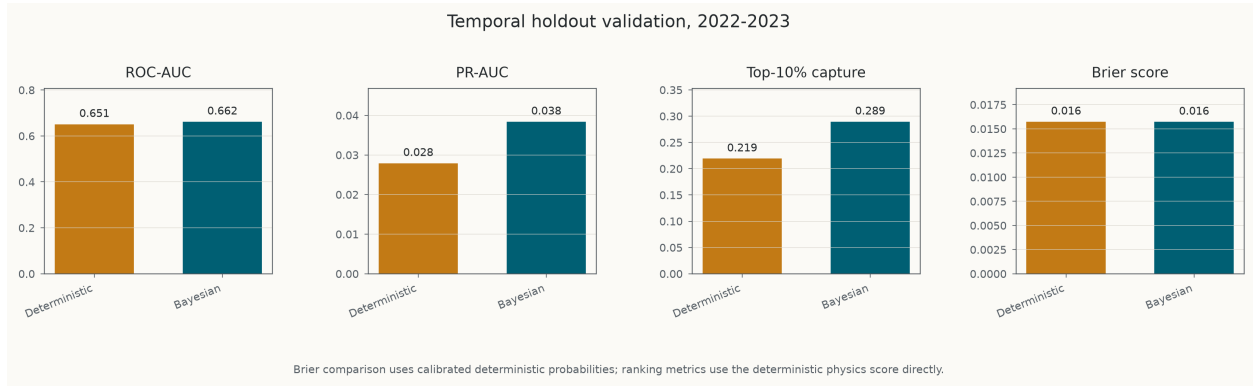


Figure 9: Temporal-holdout validation comparing deterministic and Bayesian ranking performance.

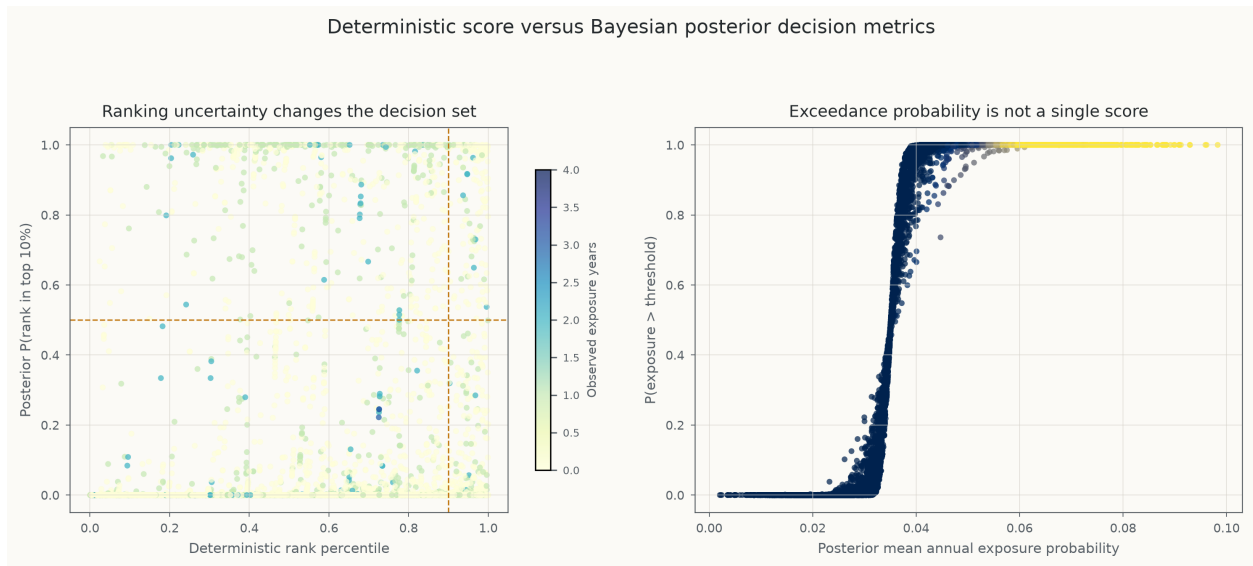


Figure 10: Deterministic score versus Bayesian posterior decision metrics. Posterior quantities expose ranking uncertainty that is hidden by a single deterministic score.

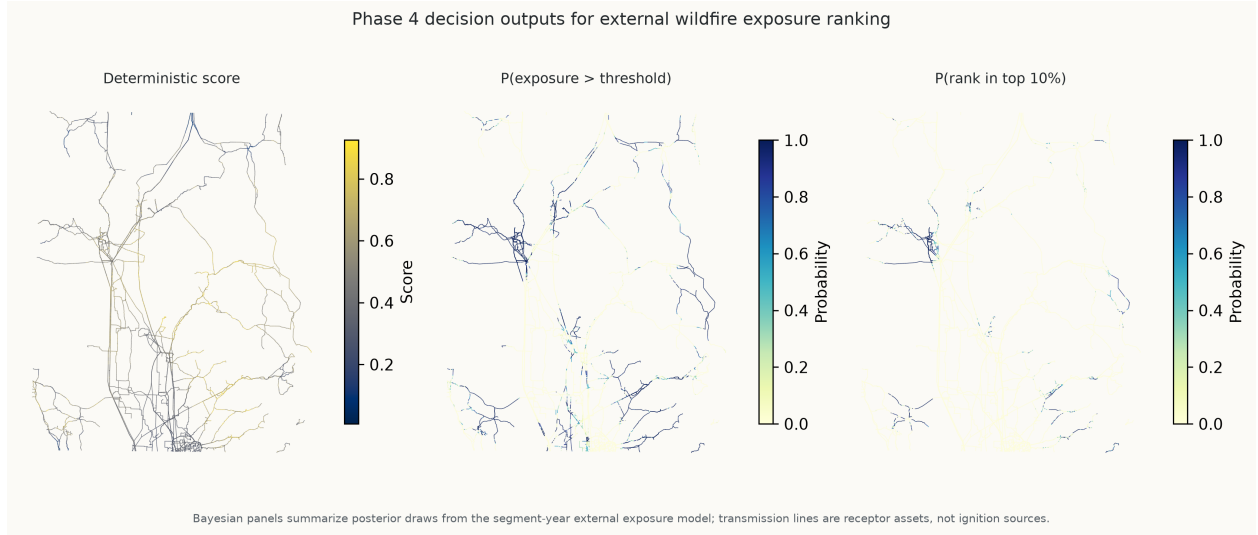


Figure 11: Spatial distribution of Bayesian posterior decision metrics for transmission-line segments.

Table 1: Main temporal-holdout comparison for 2022–2023 exogenous exposure.

Model	ROC-AUC	PR-AUC	Brier	Row top 10%	Segment top 10%
Deterministic calibrated	0.651	0.028	0.015734	0.219	0.118
Bayesian Laplace	0.662	0.038	0.015718	0.289	0.216
Stan MCMC pilot	0.662	0.038	0.015710	0.289	0.216
ML logistic L2	0.663	0.039	0.015713	0.289	0.216
Balanced extra trees	0.617	0.022	0.056091	0.180	0.104
Histogram gradient boosting	0.553	0.019	0.016647	0.132	0.149
Balanced random forest	0.568	0.018	0.034539	0.107	0.152

5.2 Feature Ablation

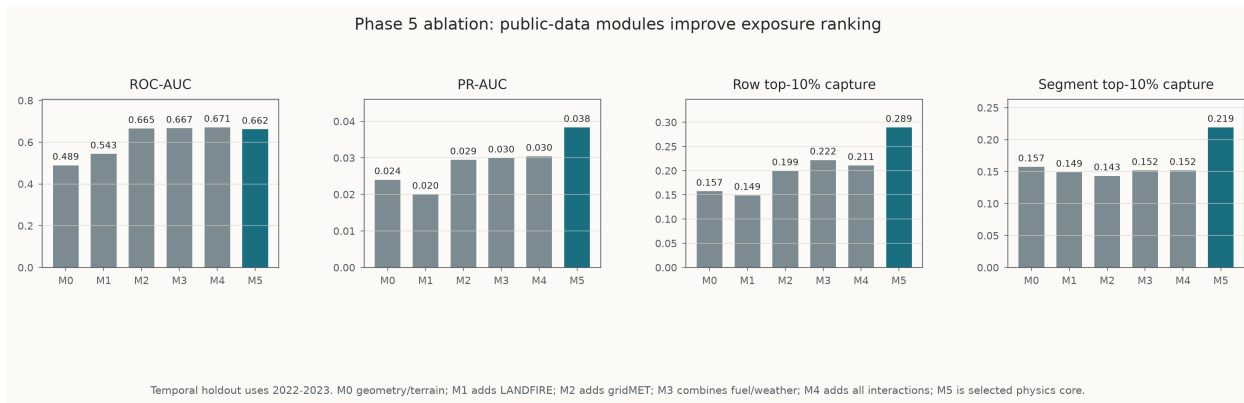


Figure 12: Feature-set ablation under temporal holdout. Dynamic gridMET variables drive much of the discrimination, while the selected physics core yields the strongest rare-event ranking concentration.

Table 2: Feature-set ablation under 2022–2023 temporal holdout.

Model	ROC-AUC	PR-AUC	Brier	Row top 10%	Segment top 10%
M0 geometry + terrain	0.489	0.024	0.016052	0.157	0.157
M1 + LANDFIRE fuel	0.543	0.020	0.016076	0.149	0.149
M2 + gridMET weather	0.665	0.029	0.015822	0.199	0.143
M3 fuel + weather	0.667	0.030	0.015821	0.222	0.152
M4 physics interaction	0.671	0.030	0.015820	0.211	0.152
M5 selected physics core	0.662	0.038	0.015723	0.289	0.219

5.3 Machine Learning and Stan Validation

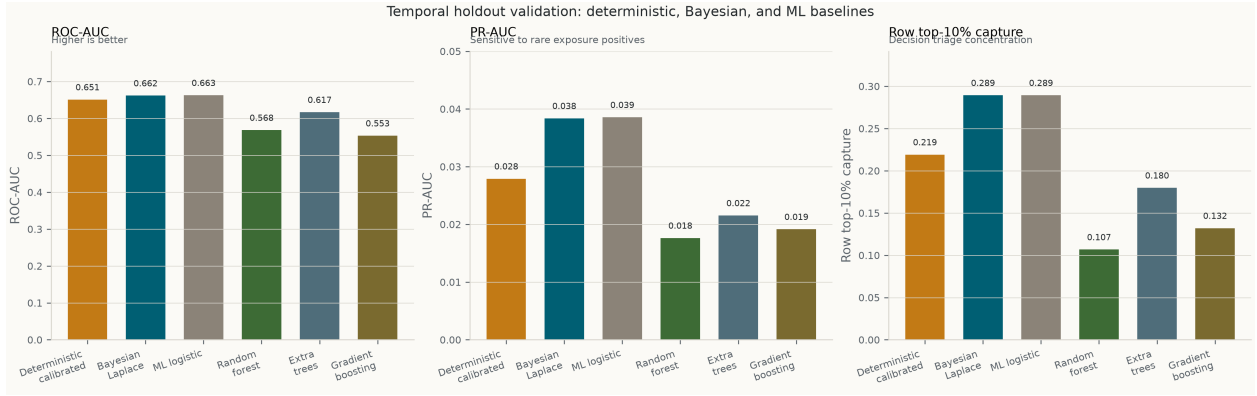


Figure 13: Deterministic, Bayesian, and machine-learning baselines under the same temporal holdout. Flexible tree-based models do not improve rare-event ranking performance over the selected physics-informed linear core.

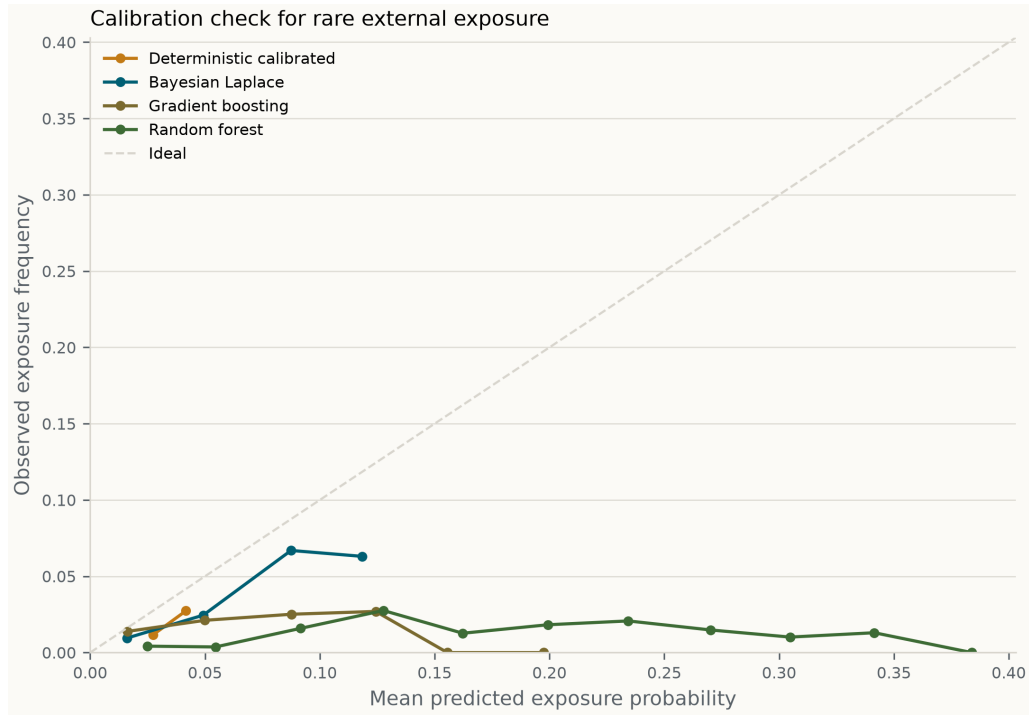


Figure 14: Calibration check for rare external exposure probabilities.

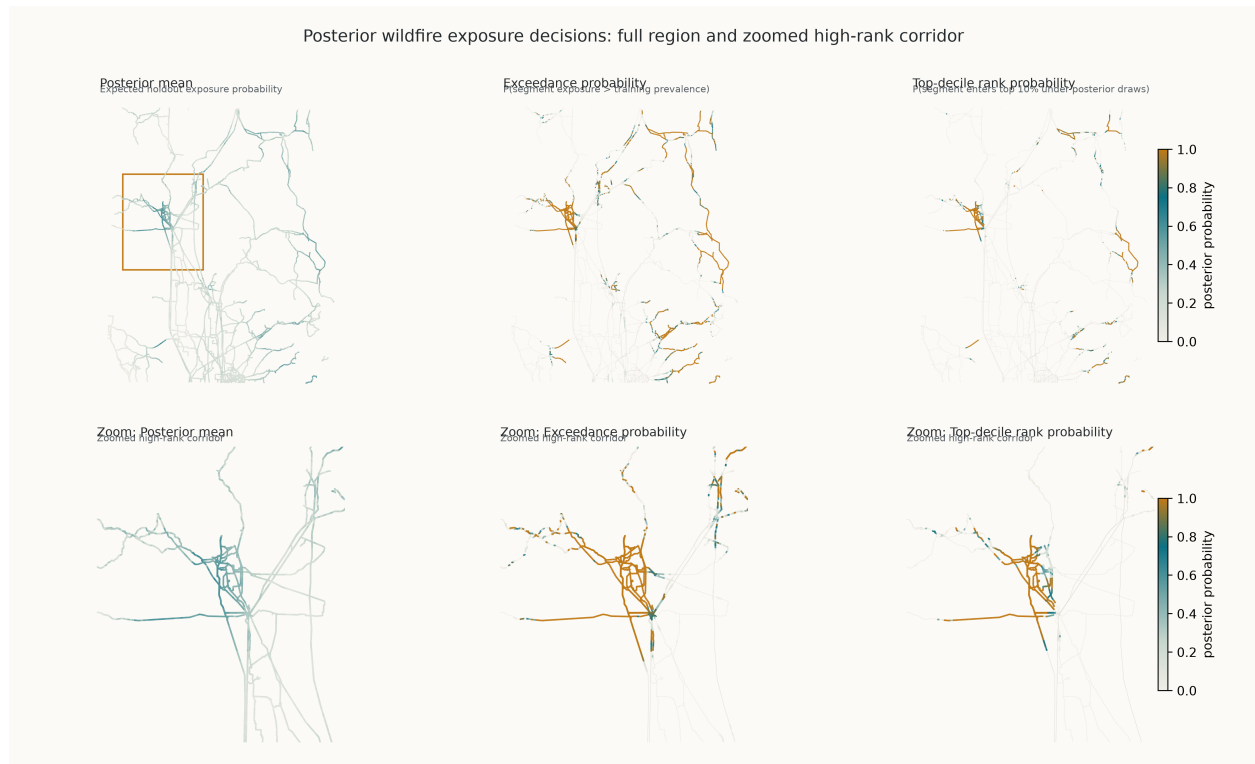


Figure 15: Faceted posterior decision maps with full-region and zoomed views. The top row locates posterior exposure patterns over the full study area; the bottom row enlarges a high-rank corridor to make local rank differences visible.

6 Discussion

The results support the central claim that Bayesian decision outputs are more appropriate than deterministic risk scores for extreme wildfire exposure triage. The deterministic score is transparent but loses information about rank uncertainty. The Bayesian model gives posterior exceedance and rank probabilities that map directly to operational decisions such as selecting a top decile of segments for inspection, hardening, or scenario review.

The ML comparison sharpens the interpretation. More flexible tree-based learners do not automatically improve temporal generalization. The selected physics-informed linear core performs best for rare-event concentration, which suggests that physically motivated feature construction is more important than model flexibility for this sparse exposure target.

6.1 Comparison with Existing Risk-Assessment Approaches

Existing wildfire decision tools commonly fall into five families. Fire-danger indices and deterministic hazard scores are transparent and operationally familiar, but they usually return a single ordered score and cannot quantify whether a segment’s rank is stable under parameter uncertainty. Generic machine learning classifiers can absorb nonlinear interactions, but in this experiment they do not dominate the physics-informed Bayesian model under a 2022–2023 temporal holdout, and their outputs require additional calibration before they can be used as exceedance or rank probabilities.

Powerline ignition and equipment-failure models answer a different question. They often depend on proprietary asset condition, protection, outage, inspection, and vegetation-management data. This study deliberately avoids that target and treats the line only as a receptor asset exposed to external fires. Fire-spread simulators and scenario models provide richer physical dynamics, but they are computationally heavier and are not automatically calibrated for segment-level posterior ranking. Bayesian hierarchical and Bayesian decision models are closest in spirit to the proposed approach. The contribution here is to combine public-data reproducibility, cause-screened exposure labels, physics-informed covariates, temporal validation, and posterior top-decile rank probabilities in a single segment-level pipeline for extreme-disaster triage.

7 Limitations

The pipeline estimates external exposure, not ignition causality. It does not include private asset-condition data, relay operations, vegetation-management records, outage logs, or equipment-failure histories. Therefore, results should not be interpreted as powerline ignition probabilities or outage probabilities. Additional validation should include spatial block holdouts, buffer sensitivity, and longer Stan MCMC diagnostics.

8 Conclusion

A reproducible public-data pipeline can rank transmission-line segments by external wildfire exposure under temporal holdout. Bayesian posterior ranking outperforms a deterministic calibrated score for rare-event decision concentration and produces decision quantities that are directly usable for extreme-disaster triage. The Stan MCMC pilot agrees with the Laplace posterior prototype, supporting the planned full Bayesian implementation.

A Reproducibility Notes

All data processing, feature construction, model fitting, validation, and figure generation are scripted in the project repository. The public-data workflow is organized by phases: asset and label construction, environmental covariate preparation, model-table construction, Bayesian ranking, ablation, ML/Stan validation, and manuscript result generation. The manuscript tables and figures are generated from those scripted outputs rather than manually transcribed.